

World Cup 2026 Simulation

A Data-Driven JavaFX Simulator and Bracket Visualization

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Why build a World Cup engine?



- **Passion:** Personal project to further enhance programming skills/data-driven insights with immense passion for World Cup football.
- **Portfolio Showcase:** Resume booster to demonstrate industry-level Java implementation through OOP and complex algorithmic sorting.



The Architecture



- **Main Engine:** 'TournamentManager' handles the core data models for group seeding, table advancement, and knockout progression.
- **UML to JavaFX:** Finalized a UML design to map out the relationships of classes within the program and frame the design based off of the diagram.



The Challenge: An expanded format



- **FIFA expansion:** 2022 held 32 teams in one country, 2026 now has 48 teams across 3 countries.
- **Knockout Algorithm:** With 32 knockout teams, the top 2 in each group automatically advance, leaving the best of the rest to be decided through tiebreakers for 3rd mathematically by their group stage performance.



Self-imposed Match Decision Handling



- **FIFA Ranking:** Each nation is represented by a FIFA rank (dispersed from 1-48 respectfully for computational efficiency) as well as win probabilities (calculated based off of their rank).
- **Goal Generation:** Utilized probabilistic distributions to favor higher-ranked teams to score more goals (though still allowing for the underdog upset in some instances).
- **Data Tracking:** Accurately handled collection of all group goals, points, and calculations for tiebreakers accordingly for the group stage to build the knockout round accordingly.



User experience with JavaFX



- **Match Details:** Beginner implementation of display for Match location and timing (would benefit from future implementation/enhancement).
- **Knockout Bracket View:** Visual knockout tree implemented with dynamic node coordinates and custom vector connections.
- **Dual Mode Option:** User can choose 'Birds-eye' or 'Selected' team view throughout the tournament progression.
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Project execution and timeline



- **Phase 1 - Project Scoping:** Initial idea rendering and data research to plan for summer project (January - May 2026).
- **Phase 2 - Implementation:** Trial and Error period of programming the project within Eclipse up until current timestamp of completion (June - August 2026).
- **Phase 3 - Visualization and Presentation:** Cleaning up all JavaFX visuals and building a professional-grade presentation (September 2026).



Future features and expansion roadmap



- **Player Design:** Implement player statistics for each team, analyze historic performances throughout their World Cup career, track all goals and allow for individual showcase awards to be displayed (Golden Boot, Golden Glove, etc.)
- **2030 World Cup goal:** Aim to include further enhancements and build towards a stronger project for the next World Cup.



Treating a personal project professionally



- **Repository Updates:** Appropriately committing project implementations to GitHub and completing a README.md document for overview of project for anyone interested.
- **Opportunity For Growth:** UML design allows for expansion on UI themes and further simulation algorithms whilst sticking to core engine within the program.





Thank You!

